

Baseline Monthly Cost

Assumptions

- Region: us-east-1
- 2x t3.micro EC2 instances running most of the time
- ASG can scale up to 4, but steady state = 2
- 1 internet-facing ALB
- Low traffic (minimal LCU usage)
- On-Demand pricing

EC2 Compute Cost

- t3.micro (On-Demand)
- ~ \$0.0104 / hr

Monthly (per instance)

- \$ 7.60

Two instances (HA baseline)

- ~ \$15.20 / mo

Scaling up to 4 instances temporarily would increase cost during those hours, but under normal conditions this remains close to the 2-instance baseline.

Application Load Balancer Cost

ALB fixed cost

- ~ \$0.0225 / hour
- \$ 16.40 / mo

LCU cost

- With low traffic and simple HTTP health checks, LCUs are usually negligible
- Expect \$1 - \$3 / mo in this setup

ALB total

- ~ \$17 - \$20 / mo

Approx Baseline

EC2 (2x t3.micro) ~ \$15

ALB ~ \$18

Total

~ \$33 / month

This architecture achieves high availability at the lowest practical cost on AWS by duplicating compute across Availability Zones, with EC2 representing the primary cost driver and Reserved or Spot capacity being the main optimization levers.